Aidan McLean

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Education

McMaster University **September 2021 – April 2026**
Bachelor of Applied Science, Honours Computer Science (Co-op) *Hamilton, ON*

Work Experience

Endnote Records **April 2025 – Present**
Co-Founder & Technical Lead *endnoterecords.ca*

- Launched the label's e-commerce platform in October 2026, taking 10 orders on day one, after building it solo as a Next.js 16 storefront and staff admin sharing one Drizzle ORM data layer over PostgreSQL in a Turborepo monorepo.
- Built a Stripe checkout that reprices every line server-side and reserves stock in one transaction using deterministically ordered row locks, with signature-verified, idempotent webhooks and a reconciliation job for missed events.
- Implemented USD and CAD pricing with no FX conversion, rendering both prices in React Server Components and selecting one with a pre-paint inline script, so product pages keep static generation with ISR and avoid layout shift.
- Enforced permission-level RBAC with Better Auth's access-control plugin on every admin route and server action, with invite-only accounts, TOTP two-factor sign-in, and an audit log kept append-only by PostgreSQL triggers.
- Validated every server action with Zod, implemented rate limits as atomic PostgreSQL upserts that hold across serverless instances, stored only salted hashes of visitor IPs, and scrubbed personal data from Sentry error reports.
- Deployed both apps on Vercel and Neon, with a GitHub Actions CI/CD pipeline that runs typecheck, lint, 142 Vitest tests, and a build against a real PostgreSQL instance, applies migrations on merge, and runs scheduled jobs.
- Moved the store to the new platform with no interruption to sales and set up branded business and customer email.

McMaster University Faculty of Engineering **June 2026 – October 2026**
Research Assistant II & Full-Stack Developer *Hamilton, ON*

- Wrote most of the TypeScript Express backend for a platform that scores and ranks battery State-of-Charge estimation models on 18+ metrics, starting on a four-person capstone team and continuing as a research assistant.
- Translated requirements from weekly planning meetings and a prioritized backlog into fixes and features across the REST API, PostgreSQL schema, and React interface, working with the project lead and development team.
- Built an asynchronous evaluation pipeline that runs each submission in a short-lived, read-only Docker container limited to 2 GB of memory, 2 CPUs, and 100 processes before parsing the scorer's JSON output.
- Secured model uploads with Multer type and size limits, per-user storage paths, filename sanitization against path traversal, and inspection of every zip entry, rejecting nested archives and disallowed file types.
- Implemented Better Auth with email verification and rate-limited sign-in, enforced roles and ownership-gated access in Express middleware, and built admin tools for bans, session revocation, and private model visibility.
- Redesigned the interface with shadcn/ui, added model and result downloads, and pushed submission and evaluation status to users by email and live over Server-Sent Events.
- Deployed the platform on Compute Canada's OpenStack cloud with Docker Compose, NGINX, PM2, and SSL/TLS, managed its pre-production environment and certificate renewal, and added structured Winston logging.

Regional Municipality of York **May 2025 – September 2025**
Software Developer Co-op Student *Newmarket, ON*

- Customized Drupal Webforms with JavaScript to implement dynamic form behaviour, validation, and improved UX.
- Built Microsoft APIM integrations to apply business logic and route form requests securely to backend services.
- Connected forms to SharePoint Lists and Power Automate to deliver secure, automated workflows for internal apps.
- Developed a TypeScript frontend application that retrieved and visualized data from an internal GIS map server.
- Documented APIs, workflows, and form logic so other developers could pick up the work without re-deriving it.

Technical Proficiencies

Languages: TypeScript/JavaScript, Python, Java, SQL, C, Bash, PowerShell
Frontend: React, Next.js (App Router, Server Components), Tailwind CSS, shadcn/ui, Base UI, responsive design
Backend & Data: Node.js, Express, REST APIs, PostgreSQL, Drizzle ORM, schema design, Zod, Better Auth, Stripe
Infrastructure & DevOps: Docker, Docker Compose, NGINX, PM2, Linux, OpenStack, Vercel, Neon, GitHub Actions
Testing & Tooling: Vitest, Git, pnpm, Turborepo, ESLint, Prettier, Sentry, Agile/Scrum
Enterprise Platforms: Drupal, Microsoft APIM, Power Automate, SharePoint, Microsoft Entra ID, GIS services